

Archived Analysis: Household Inflation Burden

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October 9, 2026

This standalone copy preserves the inflation-burden analysis, which has been restored to the main manuscript with an explicit interpretation caveat. Net household income is used both to assign income quintiles and as the denominator of the burden measure. The resulting Q1–Q5 gradient can therefore partly reflect a mechanical expenditure-to-income gradient rather than heterogeneous inflation exposure. Before interpreting the gradient as an inflation-exposure effect, the analysis should:

1. decompose the gap into heterogeneous inflation and heterogeneous expenditure-to-income ratios;
2. report both means and medians;
3. test alternative winsorization thresholds;
4. rank households using equivalised income; and
5. calculate the burden under a common inflation rate to isolate the mechanical denominator channel.

1 Inflation Burden

Our indicators measure differences in inflation exposure across household groups, but they do not directly express the cost of inflation relative to household resources. We therefore construct the burden from HBS microdata rather than combining group-level inflation with an external aggregate consumption-to-income ratio. For household i in country c , let X_{ic} denote consumption expenditure on the non-rent products matched to HICP, Y_{ic} net household income, and $\bar{\pi}_{ic,y}$ the household’s average annual inflation rate in year y , computed with the RAS-calibrated weights used in the main manuscript. The raw household-level burden is

$$\mathcal{B}_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}. \quad (1)$$

Because $\bar{\pi}_{ic,y}$ is measured in percent, $\mathcal{B}_{ic,y}^{\text{raw}}$ is measured in points of net income. The indicator allows both individual inflation exposure and the consumption-to-income ratio to vary across households. The expenditure numerator and inflation index cover the same matched non-rent products. This fixed-resource exposure indicator is not compensating variation: it holds 2020 expenditure and income fixed and allows neither substitution nor income adjustment.

Very small reported incomes can generate mechanically extreme expenditure-to-income ratios. We therefore winsorize $100X_{ic}/Y_{ic}$ at the survey-weighted 99th percentile within each

country, while retaining the underlying household inflation rate. Denoting the resulting ratio by $\tilde{\kappa}_{ic}$, the burden used in the estimates and its aggregation for national income quintile q are

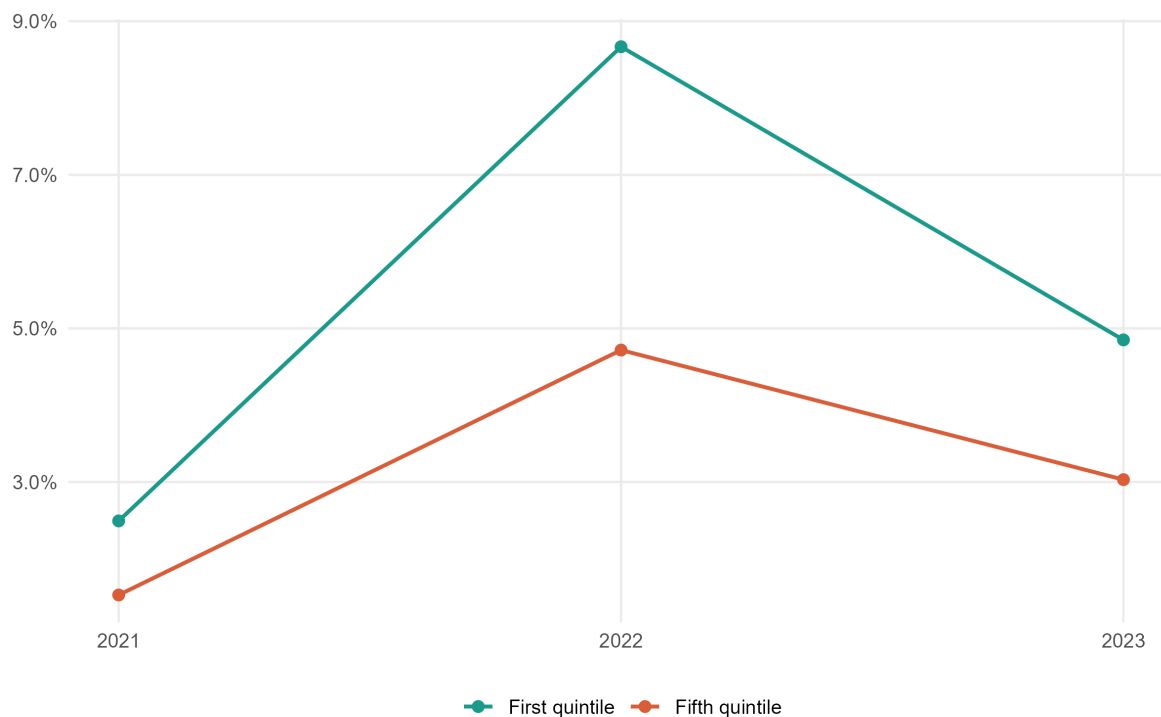
$$\mathcal{B}_{ic,y} = \bar{\pi}_{ic,y} \frac{\tilde{\kappa}_{ic}}{100}, \quad \mathcal{B}_{q,y} = \frac{\sum_c \sum_{i \in q(c)} a_{ic} \mathcal{B}_{ic,y}}{\sum_c \sum_{i \in q(c)} a_{ic}}. \quad (2)$$

Here a_{ic} is the HBS survey weight, and $q(c)$ indicates that total-net-income quintiles are defined separately within each country before households are pooled. The survey weights expand the microdata to national household populations and therefore determine the euro-area aggregation. The 2020 HBS basket, income, and expenditure observations are held fixed, while household-specific inflation is calculated for 2021–2023. For each household-year, the inflation rate is computed over the matched non-rent expenditure categories and the HBS–HICP product weights are renormalized to one. We impose no minimum matched-expenditure threshold.

Higher-income households also generally have greater capacity to absorb a price shock by reducing discretionary expenditure or substituting toward cheaper varieties. The main manuscript’s appendix documents the corresponding difference in basket composition.

Figure 1 uses household indices and matched non-rent expenditure. The first quintile faces a burden of 2.5 percent of net income in 2021, 8.7 percent in 2022, and 4.8 percent in 2023, compared with 1.5, 4.7, and 3.0 percent for the fifth quintile. The Q1–Q5 gap is 1.0, 4.0, and 1.8 points, respectively, calculated before rounding. The income and expenditure filters retain positive-income households; national quintiles are assigned before these burden-specific exclusions.

Figure 1: Household-level inflation burden in the euro area, by income quintile (in % of net income)



Notes: Annual household inflation times matched non-rent expenditure divided by net income; ratios are winsorized at the national survey-weighted 99th percentile. Survey-weighted means within national total-income quintiles, 18 countries excluding Italy and Portugal. Fixed 2020 resources; this is an exposure measure, not a welfare estimate.

Sources: HBS microdata, Eurostat HICP, authors' calculations.